

ACTIVITY-4

1) Input values { 4871, 1323, 6173, 4199, 4344, 9679, 1989 } and a hash function $h(x) = x \text{ (mod } 10)$, show the resulting

a) ~~separate~~ separate chaining
hash table

Table size = 10 - ^e separate chaining

key	$h(x)$
4871	1
1323	3
6173	3
4199	9
4344	4
9679	9
1989	9

collision
occurs at
3, 9

index	chain
0	-
1	4871
2	-
3	1323 → 6173
4	4344
5	-
6	-
7	-
8	-
9	4199 → 9679 → 1989

b) open addressing $\rightarrow$ Linear Probing
formula

$$h(x) = (h(x) + 1) \bmod 10$$

$$(6173 + 1) \cdot 10 = 4 \text{ occupied}$$

$$(6174 + 1) \cdot 10 = 5 \text{ free}$$

$$(9679 + 1) \cdot 10 = 0 \text{ free}$$

$$(1989 + 1) \cdot 10 = 0 \text{ occupied}$$

$$(1990 + 1) \cdot 10 = 1 \text{ occupied}$$

$$(1991 + 1) \cdot 10 = 2 \text{ free}$$

final hash table

index	value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

(iii) open addressing - Quadratic probing
formula :

$$h_i(x) = (h(x) + i^2) \bmod 10$$

4371

1 $\rightarrow$ index 1

1323

3 $\rightarrow$ index 3

6173

3 occupied

$$i = 1, 3 + 1^2 = 4$$

index 4.

4199

9 $\rightarrow$ index 9

41344

4 occupied

$$4 + 1^2 = 5 \rightarrow \text{index 5}$$

9679

9 occupied

$$9 + 1^2 = 0$$

$\rightarrow$ index 0

1989

$i = 1$; 0 occupied

$$i = 2 ; (9 + 4) \bmod 10 = 3 \text{ occupied}$$

$$i = 3 ; (9 + 9) \bmod 10 = 8 \text{ index 8}$$

index	Value
0	9679
1	4371
2	-
3	1323
4	6173
5	4344
6	-
7	-
8	1989
9	4199

double hashing

$$h_1(x) = x \mod 10$$

and

$$h_2(x) = 7 - (x \mod 7)$$

The Probing formula is

$$h_i(x) = (h_1(x) + i \times h_2(x))$$

mod 10

calculate $h_2(x)$

key	$h_1(x)$	$x \mod 7$	$h_2(x) = 7 - x \mod 7$
4371	1	5	2
1323	3	2	5
6173	3	6	1
4199	9	3	4
4344	4	4	3
9679	9	5	2

1989

9

1

6

insertion :

4371

 $h_1 = 1 \rightarrow \text{index } 1$

1323

 $h_1 = 3 \rightarrow \text{index } 3$

6173

 $h_1 = 3$ occupiedStep Size $h_2 = 1$ $3 + 1 = 4$

index = 4

4199 :

 $h_1 = 9 \rightarrow \text{index } 9$

4344

 $h_1 = 4$ occupied $h_1 = 3$

⋮

After
19/8/20

1989 $\rightarrow$ cannot be inserted
using double -
hashing setup

Final result
index value

0

1

4371

2

-

3

1323

4

6173

5

9879

6

-

7

4344

8

-

9

4199